Skill 11

Use Statistical Sense

What Do the Numbers Say?

160 Items	**1-5** Levels	**Read numbers with the right context.** Objective

Essential question. What does this number mean in context?

This PDF is designed for classroom use. It gives teachers a rigorous introduction to the skill, a repeatable solving routine, likely student confusions, a worked example, and the full XiXteen item set with choice-level feedback.

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Skill 11

Use Statistical Sense

What Do the Numbers Say?

Read numbers with the right context. Objective	**160** Items	**Levels 1-5** Difficulty bands

Essential question	What does this number mean in context?
Why it matters	Numbers can clarify, but they can also mislead when people ignore denominators, baselines, rates, averages, or the difference between percent and percentage points.
Conceptual core	Statistical sense keeps numbers connected to their denominators, baselines, units, and comparisons. Students should learn that a precise number can still be misleading if the reference point is missing. The central habits are asking compared with what, out of how many, measured how, and over what time period. The goal is not advanced math; it is disciplined interpretation.
Student method	Ask what is being counted, what it is being compared with, and what the denominator is. Never let a large raw number outrun the size of the group it came from.
Analogy	Statistics are like a scale. If you do not know the units or the zero point, the reading can look precise while meaning very little.
Assessment focus	A good answer identifies the missing or correct numerical context: baseline, denominator, comparison group, unit, or time period.

Teaching Sequence

1. Identify the numerator and denominator behind a rate.

2. Ask what the number is being compared with.

3. Separate percent change from percentage-point change.

4. Check whether averages, totals, or selected examples hide variation.

Common Confusions To Watch For

1. Students may think a bigger raw count always means a bigger rate.

2. Students may treat a percentage change as clear without knowing the starting value.

3. Students may ignore whether the comparison groups are the same size.

Teacher Prompts

1. Out of how many?

2. Compared with what baseline?

3. Is this a total, rate, average, percentage, or percentage-point change?

Worked Example

Scenario	A clinic says missed appointments fell by 20 percent after reminder texts, from 100 missed visits per month to 80.
Teacher think-aloud	The 20 percent has a reference point: 20 fewer missed visits compared with the earlier 100. Without the before number, the change would be harder to judge.
Student takeaway	Numbers become meaningful when the comparison and denominator are visible.

Classroom extension. Have students convert every statistic in a news paragraph into a sentence that names the denominator and comparison.

Item Bank

Items are grouped by level. For each item, read the prompt, predict the answer, choose, and then use the feedback and process commentary to examine the reasoning path.

Level 1

Level 1 establishes the clean version of the skill: the cue is direct, the distractors are visible, and the main task is to learn the contrast.

statistical_sense-001 | Level 1 | Answer D

Prompt. In a study about evening bus service for late-shift workers, the tracked outcome is late-shift commute complaints. Group A recorded 5 outcome events in 100 observations. Group B recorded 7 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group B, because 7 events is more than 5 events, so the raw count must decide the rate. choice	**D.** Group A, because 5/100 is a higher rate than 7/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, transportation, d1

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-002 | Level 1 | Answer C

Prompt. In a study about classroom phone lockers, the tracked outcome is classroom disruption reports. Group A recorded 8 outcome events in 300 observations. Group B recorded 6 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 8 events is more than 6 events, so the raw count must decide the rate. choice	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group B, because 6/100 is a higher rate than 8/300; compare rates, not just counts. correct	**D.** Group A, because 300 observations is the larger group, so it must have the higher rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, education, d1

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

statistical_sense-003 | Level 1 | Answer B

Prompt. In a study about medication reminder texts for older patients, the tracked outcome is missed-dose reports. Group A recorded 7 outcome events in 100 observations. Group B recorded 9 outcome events in 300 observations. Which group had the higher rate?

A. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**B.** Group A, because 7/100 is a higher rate than 9/300; compare rates, not just counts. correct
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** Group B, because 9 events is more than 7 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, health, d1

A feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-004 | Level 1 | Answer A

Prompt. In a study about a four-day workweek test for software engineers, the tracked outcome is staff resignations. Group A recorded 10 outcome events in 300 observations. Group B recorded 8 outcome events in 100 observations. Which group had the higher rate?

A. Group B, because 8/100 is a higher rate than 10/300; compare rates, not just counts. correct	**B.** Group A, because 10 events is more than 8 events, so the raw count must decide the rate. choice
C. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**D.** The raw counts are close enough that dividing by group size is unnecessary. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, workplace, d1

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-005 | Level 1 | Answer D

Prompt. In a study about larger grocery unit-price labels, the tracked outcome is shopper overpayment complaints. Group A recorded 9 outcome events in 100 observations. Group B recorded 11 outcome events in 300 observations. Which group had the higher rate?

A. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**B.** Group B, because 11 events is more than 9 events, so the raw count must decide the rate. choice
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** Group A, because 9/100 is a higher rate than 11/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, retail, d1

A feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-006 | Level 1 | Answer C

Prompt. In a study about late fines on children's books, the tracked outcome is library card renewals. Group A recorded 12 outcome events in 300 observations. Group B recorded 10 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice
C. Group B, because 10/100 is a higher rate than 12/300; compare rates, not just counts. correct	**D.** Group A, because 12 events is more than 10 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, public services, d1

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-007 | Level 1 | Answer B

Prompt. In a study about two-factor login for site administrators, the tracked outcome is admin account takeovers. Group A recorded 11 outcome events in 100 observations. Group B recorded 13 outcome events in 300 observations. Which group had the higher rate?

A. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**B.** Group A, because 11/100 is a higher rate than 13/300; compare rates, not just counts. correct
C. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** Group B, because 13 events is more than 11 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, technology, d1

A feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-008 | Level 1 | Answer A

Prompt. In a study about brighter streetlights on Maple Street, the tracked outcome is reported thefts. Group A recorded 14 outcome events in 300 observations. Group B recorded 12 outcome events in 100 observations. Which group had the higher rate?

A. Group B, because 12/100 is a higher rate than 14/300; compare rates, not just counts. correct	**B.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice
C. Group A, because 14 events is more than 12 events, so the raw count must decide the rate. choice	**D.** The raw counts are close enough that dividing by group size is unnecessary. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, public safety, d1

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-009 | Level 1 | Answer D

Prompt. In a study about peer tutoring for first-year statistics students, the tracked outcome is course pass rates. Group A recorded 4 outcome events in 100 observations. Group B recorded 6 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 6 events is more than 4 events, so the raw count must decide the rate. choice	**B.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** Group A, because 4/100 is a higher rate than 6/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, education, d1

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-010 | Level 1 | Answer C

Prompt. In a study about calorie labels on restaurant menus, the tracked outcome is dessert orders. Group A recorded 7 outcome events in 300 observations. Group B recorded 5 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 7 events is more than 5 events, so the raw count must decide the rate. choice	**B.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice
C. Group B, because 5/100 is a higher rate than 7/300; compare rates, not just counts. correct	**D.** The raw counts are close enough that dividing by group size is unnecessary. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, nutrition, d1

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-011 | Level 1 | Answer C

Prompt. In a study about bank overdraft warning texts, the tracked outcome is overdraft fees. Group A recorded 6 outcome events in 100 observations. Group B recorded 8 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 8 events is more than 6 events, so the raw count must decide the rate. choice	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group A, because 6/100 is a higher rate than 8/300; compare rates, not just counts. correct	**D.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, finance, d1

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

statistical_sense-012 | Level 1 | Answer D

Prompt. In a study about timed-entry passes for weekend park trails, the tracked outcome is trail crowding complaints. Group A recorded 9 outcome events in 300 observations. Group B recorded 7 outcome events in 100 observations. Which group had the higher rate?

A. The raw counts are close enough that dividing by group size is unnecessary. choice	**B.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice
C. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** Group B, because 7/100 is a higher rate than 9/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, environment, d1

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-013 | Level 1 | Answer A

Prompt. In a study about free Friday museum admission, the tracked outcome is family visits. Group A recorded 8 outcome events in 100 observations. Group B recorded 10 outcome events in 300 observations. Which group had the higher rate?

A. Group A, because 8/100 is a higher rate than 10/300; compare rates, not just counts. correct	**B.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice
C. Group B, because 10 events is more than 8 events, so the raw count must decide the rate. choice	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, culture, d1

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-014 | Level 1 | Answer B

Prompt. In a study about youth soccer concussion checks, the tracked outcome is repeat head injuries. Group A recorded 11 outcome events in 300 observations. Group B recorded 9 outcome events in 100 observations. Which group had the higher rate?

A. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**B.** Group B, because 9/100 is a higher rate than 11/300; compare rates, not just counts. correct
C. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** Group A, because 11 events is more than 9 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, sports, d1

A feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-015 | Level 1 | Answer A

Prompt. In a study about same-day donation receipts, the tracked outcome is donor retention. Group A recorded 10 outcome events in 100 observations. Group B recorded 12 outcome events in 300 observations. Which group had the higher rate?

A. Group A, because 10/100 is a higher rate than 12/300; compare rates, not just counts. correct	**B.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, nonprofit, d1

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-016 | Level 1 | Answer C

Prompt. In a study about warehouse ergonomic training, the tracked outcome is back-injury claims. Group A recorded 13 outcome events in 300 observations. Group B recorded 11 outcome events in 100 observations. Which group had the higher rate?

A. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**B.** Group A, because 300 observations is the larger group, so it must have the higher rate. choice
C. Group B, because 11/100 is a higher rate than 13/300; compare rates, not just counts. correct	**D.** Group A, because 13 events is more than 11 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, workplace safety, d1

A feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

B feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-017 | Level 1 | Answer A

Prompt. In a study about apartment-lobby compost bins, the tracked outcome is landfill waste from participating buildings. Group A recorded 12 outcome events in 100 observations. Group B recorded 14 outcome events in 300 observations. Which group had the higher rate?

A. Group A, because 12/100 is a higher rate than 14/300; compare rates, not just counts. correct	**B.** Group B, because 14 events is more than 12 events, so the raw count must decide the rate. choice
C. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** The raw counts are close enough that dividing by group size is unnecessary. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, environment, d1

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-018 | Level 1 | Answer C

Prompt. In a study about private-by-default app profiles, the tracked outcome is publicly visible profiles. Group A recorded 6 outcome events in 300 observations. Group B recorded 4 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group B, because 4/100 is a higher rate than 6/300; compare rates, not just counts. correct	**D.** Group A, because 6 events is more than 4 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, technology, d1

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-019 | Level 1 | Answer A

Prompt. In a study about same-day clinic appointment slots, the tracked outcome is patient no-show rates. Group A recorded 5 outcome events in 100 observations. Group B recorded 7 outcome events in 300 observations. Which group had the higher rate?

A. Group A, because 5/100 is a higher rate than 7/300; compare rates, not just counts. correct	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**D.** Group B, because 7 events is more than 5 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, health, d1

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-020 | Level 1 | Answer C

Prompt. In a study about seat-zone airline boarding, the tracked outcome is average boarding time. Group A recorded 8 outcome events in 300 observations. Group B recorded 6 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** Group A, because 8 events is more than 6 events, so the raw count must decide the rate. choice
C. Group B, because 6/100 is a higher rate than 8/300; compare rates, not just counts. correct	**D.** The raw counts are close enough that dividing by group size is unnecessary. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, travel, d1

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-021 | Level 1 | Answer D

Prompt. In a study about an online apartment repair portal, the tracked outcome is unresolved repair requests. Group A recorded 7 outcome events in 100 observations. Group B recorded 9 outcome events in 300 observations. Which group had the higher rate?

A. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**B.** Group B, because 9 events is more than 7 events, so the raw count must decide the rate. choice
C. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** Group A, because 7/100 is a higher rate than 9/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, housing, d1

A feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-022 | Level 1 | Answer B

Prompt. In a study about fruit placement in a school cafeteria line, the tracked outcome is fruit purchases. Group A recorded 10 outcome events in 300 observations. Group B recorded 8 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 10 events is more than 8 events, so the raw count must decide the rate. choice	**B.** Group B, because 8/100 is a higher rate than 10/300; compare rates, not just counts. correct
C. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**D.** Group A, because 300 observations is the larger group, so it must have the higher rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, food, d1

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

D feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

statistical_sense-023 | Level 1 | Answer D

Prompt. In a study about weekly call-center script coaching, the tracked outcome is customer complaint rates. Group A recorded 9 outcome events in 100 observations. Group B recorded 11 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 11 events is more than 9 events, so the raw count must decide the rate. choice	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** Group A, because 9/100 is a higher rate than 11/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, customer support, d1

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-024 | Level 1 | Answer C

Prompt. In a study about household water-leak alerts, the tracked outcome is average household water use. Group A recorded 12 outcome events in 300 observations. Group B recorded 10 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 12 events is more than 10 events, so the raw count must decide the rate. choice	**B.** Group A, because 300 observations is the larger group, so it must have the higher rate. choice
C. Group B, because 10/100 is a higher rate than 12/300; compare rates, not just counts. correct	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, utilities, d1

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-025 | Level 1 | Answer D

Prompt. In a study about body-camera activation reminders, the tracked outcome is missing-camera-footage reports. Group A recorded 11 outcome events in 100 observations. Group B recorded 13 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice
C. Group B, because 13 events is more than 11 events, so the raw count must decide the rate. choice	**D.** Group A, because 11/100 is a higher rate than 13/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, governance, d1

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-026 | Level 1 | Answer B

Prompt. In a study about county mail-ballot tracking, the tracked outcome is phone calls asking ballot status. Group A recorded 14 outcome events in 300 observations. Group B recorded 12 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** Group B, because 12/100 is a higher rate than 14/300; compare rates, not just counts. correct
C. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**D.** Group A, because 14 events is more than 12 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, election administration, d1

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-027 | Level 1 | Answer D

Prompt. In a study about language-learning spaced-review prompts, the tracked outcome is vocabulary retention quiz scores. Group A recorded 4 outcome events in 100 observations. Group B recorded 6 outcome events in 300 observations. Which group had the higher rate?

A. The raw counts are close enough that dividing by group size is unnecessary. choice	**B.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice
C. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**D.** Group A, because 4/100 is a higher rate than 6/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, learning, d1

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

C feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-028 | Level 1 | Answer B

Prompt. In a study about a beginner gym orientation, the tracked outcome is new-member cancellations. Group A recorded 7 outcome events in 300 observations. Group B recorded 5 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 7 events is more than 5 events, so the raw count must decide the rate. choice	**B.** Group B, because 5/100 is a higher rate than 7/300; compare rates, not just counts. correct
C. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, fitness, d1

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-029 | Level 1 | Answer B

Prompt. In a study about safe-driving feedback for policyholders, the tracked outcome is reported minor crashes. Group A recorded 6 outcome events in 100 observations. Group B recorded 8 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** Group A, because 6/100 is a higher rate than 8/300; compare rates, not just counts. correct
C. Group B, because 8 events is more than 6 events, so the raw count must decide the rate. choice	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, insurance, d1

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-030 | Level 1 | Answer A

Prompt. In a study about top-of-article correction boxes, the tracked outcome is reader trust ratings. Group A recorded 9 outcome events in 300 observations. Group B recorded 7 outcome events in 100 observations. Which group had the higher rate?

A. Group B, because 7/100 is a higher rate than 9/300; compare rates, not just counts. correct	**B.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice
C. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** Group A, because 9 events is more than 7 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, media, d1

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-031 | Level 1 | Answer A

Prompt. In a study about clearer farmers market price signs, the tracked outcome is price disputes. Group A recorded 8 outcome events in 100 observations. Group B recorded 10 outcome events in 300 observations. Which group had the higher rate?

A. Group A, because 8/100 is a higher rate than 10/300; compare rates, not just counts. correct	**B.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice
C. Group B, because 10 events is more than 8 events, so the raw count must decide the rate. choice	**D.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, local commerce, d1

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

statistical_sense-032 | Level 1 | Answer B

Prompt. In a study about hotel quiet-hours text reminders, the tracked outcome is noise complaints from guest rooms. Group A recorded 11 outcome events in 300 observations. Group B recorded 9 outcome events in 100 observations. Which group had the higher rate?

A. The raw counts are close enough that dividing by group size is unnecessary. choice	**B.** Group B, because 9/100 is a higher rate than 11/300; compare rates, not just counts. correct
C. Group A, because 11 events is more than 9 events, so the raw count must decide the rate. choice	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, hospitality, d1

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

Level 2

Level 2 keeps the same skill but adds more realistic wording and more tempting distractors.

statistical_sense-033 | Level 2 | Answer D

Prompt. In a study about evening bus service for late-shift workers, the tracked outcome is late-shift commute complaints. Group A recorded 8 outcome events in 300 observations. Group B recorded 6 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 8 events is more than 6 events, so the raw count must decide the rate. choice	**B.** Group A, because 300 observations is the larger group, so it must have the higher rate. choice
C. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**D.** Group B, because 6/100 is a higher rate than 8/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, transportation, d2

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

C feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-034 | Level 2 | Answer A

Prompt. In a study about classroom phone lockers, the tracked outcome is classroom disruption reports. Group A recorded 7 outcome events in 100 observations. Group B recorded 9 outcome events in 300 observations. Which group had the higher rate?

A. Group A, because 7/100 is a higher rate than 9/300; compare rates, not just counts. correct	**B.** Group B, because 9 events is more than 7 events, so the raw count must decide the rate. choice
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, education, d2

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

statistical_sense-035 | Level 2 | Answer B

Prompt. In a study about medication reminder texts for older patients, the tracked outcome is missed-dose reports. Group A recorded 10 outcome events in 300 observations. Group B recorded 8 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** Group B, because 8/100 is a higher rate than 10/300; compare rates, not just counts. correct
C. Group A, because 10 events is more than 8 events, so the raw count must decide the rate. choice	**D.** The raw counts are close enough that dividing by group size is unnecessary. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, health, d2

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-036 | Level 2 | Answer C

Prompt. In a study about a four-day workweek test for software engineers, the tracked outcome is staff resignations. Group A recorded 9 outcome events in 100 observations. Group B recorded 11 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 11 events is more than 9 events, so the raw count must decide the rate. choice	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group A, because 9/100 is a higher rate than 11/300; compare rates, not just counts. correct	**D.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, workplace, d2

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

statistical_sense-037 | Level 2 | Answer D

Prompt. In a study about larger grocery unit-price labels, the tracked outcome is shopper overpayment complaints. Group A recorded 12 outcome events in 300 observations. Group B recorded 10 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**D.** Group B, because 10/100 is a higher rate than 12/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, retail, d2

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-038 | Level 2 | Answer A

Prompt. In a study about late fines on children's books, the tracked outcome is library card renewals. Group A recorded 11 outcome events in 100 observations. Group B recorded 13 outcome events in 300 observations. Which group had the higher rate?

A. Group A, because 11/100 is a higher rate than 13/300; compare rates, not just counts. correct	**B.** Group B, because 13 events is more than 11 events, so the raw count must decide the rate. choice
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, public services, d2

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

statistical_sense-039 | Level 2 | Answer B

Prompt. In a study about two-factor login for site administrators, the tracked outcome is admin account takeovers. Group A recorded 14 outcome events in 300 observations. Group B recorded 12 outcome events in 100 observations. Which group had the higher rate?

A. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**B.** Group B, because 12/100 is a higher rate than 14/300; compare rates, not just counts. correct
C. Group A, because 14 events is more than 12 events, so the raw count must decide the rate. choice	**D.** Group A, because 300 observations is the larger group, so it must have the higher rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, technology, d2

A feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

statistical_sense-040 | Level 2 | Answer C

Prompt. In a study about brighter streetlights on Maple Street, the tracked outcome is reported thefts. Group A recorded 4 outcome events in 100 observations. Group B recorded 6 outcome events in 300 observations. Which group had the higher rate?

A. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**B.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice
C. Group A, because 4/100 is a higher rate than 6/300; compare rates, not just counts. correct	**D.** Group B, because 6 events is more than 4 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, public safety, d2

A feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

B feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-041 | Level 2 | Answer B

Prompt. In a study about peer tutoring for first-year statistics students, the tracked outcome is course pass rates. Group A recorded 7 outcome events in 300 observations. Group B recorded 5 outcome events in 100 observations. Which group had the higher rate?

A. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**B.** Group B, because 5/100 is a higher rate than 7/300; compare rates, not just counts. correct
C. Group A, because 7 events is more than 5 events, so the raw count must decide the rate. choice	**D.** The raw counts are close enough that dividing by group size is unnecessary. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, education, d2

A feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-042 | Level 2 | Answer C

Prompt. In a study about calorie labels on restaurant menus, the tracked outcome is dessert orders. Group A recorded 6 outcome events in 100 observations. Group B recorded 8 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group A, because 6/100 is a higher rate than 8/300; compare rates, not just counts. correct	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, nutrition, d2

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-043 | Level 2 | Answer C

Prompt. In a study about bank overdraft warning texts, the tracked outcome is overdraft fees. Group A recorded 9 outcome events in 300 observations. Group B recorded 7 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** Group A, because 9 events is more than 7 events, so the raw count must decide the rate. choice
C. Group B, because 7/100 is a higher rate than 9/300; compare rates, not just counts. correct	**D.** The raw counts are close enough that dividing by group size is unnecessary. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, finance, d2

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-044 | Level 2 | Answer B

Prompt. In a study about timed-entry passes for weekend park trails, the tracked outcome is trail crowding complaints. Group A recorded 8 outcome events in 100 observations. Group B recorded 10 outcome events in 300 observations. Which group had the higher rate?

A. The raw counts are close enough that dividing by group size is unnecessary. choice	**B.** Group A, because 8/100 is a higher rate than 10/300; compare rates, not just counts. correct
C. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** Group B, because 10 events is more than 8 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, environment, d2

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-045 | Level 2 | Answer A

Prompt. In a study about free Friday museum admission, the tracked outcome is family visits. Group A recorded 11 outcome events in 300 observations. Group B recorded 9 outcome events in 100 observations. Which group had the higher rate?

A. Group B, because 9/100 is a higher rate than 11/300; compare rates, not just counts. correct	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**D.** Group A, because 11 events is more than 9 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, culture, d2

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-046 | Level 2 | Answer D

Prompt. In a study about youth soccer concussion checks, the tracked outcome is repeat head injuries. Group A recorded 10 outcome events in 100 observations. Group B recorded 12 outcome events in 300 observations. Which group had the higher rate?

A. The raw counts are close enough that dividing by group size is unnecessary. choice	**B.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice
C. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** Group A, because 10/100 is a higher rate than 12/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, sports, d2

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-047 | Level 2 | Answer C

Prompt. In a study about same-day donation receipts, the tracked outcome is donor retention. Group A recorded 13 outcome events in 300 observations. Group B recorded 11 outcome events in 100 observations. Which group had the higher rate?

A. The raw counts are close enough that dividing by group size is unnecessary. choice	**B.** Group A, because 13 events is more than 11 events, so the raw count must decide the rate. choice
C. Group B, because 11/100 is a higher rate than 13/300; compare rates, not just counts. correct	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, nonprofit, d2

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-048 | Level 2 | Answer B

Prompt. In a study about warehouse ergonomic training, the tracked outcome is back-injury claims. Group A recorded 12 outcome events in 100 observations. Group B recorded 14 outcome events in 300 observations. Which group had the higher rate?

A. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**B.** Group A, because 12/100 is a higher rate than 14/300; compare rates, not just counts. correct
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, workplace safety, d2

A feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

statistical_sense-049 | Level 2 | Answer A

Prompt. In a study about apartment-lobby compost bins, the tracked outcome is landfill waste from participating buildings. Group A recorded 6 outcome events in 300 observations. Group B recorded 4 outcome events in 100 observations. Which group had the higher rate?

A. Group B, because 4/100 is a higher rate than 6/300; compare rates, not just counts. correct	**B.** Group A, because 6 events is more than 4 events, so the raw count must decide the rate. choice
C. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**D.** Group A, because 300 observations is the larger group, so it must have the higher rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, environment, d2

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

D feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

statistical_sense-050 | Level 2 | Answer D

Prompt. In a study about private-by-default app profiles, the tracked outcome is publicly visible profiles. Group A recorded 5 outcome events in 100 observations. Group B recorded 7 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 7 events is more than 5 events, so the raw count must decide the rate. choice	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** Group A, because 5/100 is a higher rate than 7/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, technology, d2

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-051 | Level 2 | Answer A

Prompt. In a study about same-day clinic appointment slots, the tracked outcome is patient no-show rates. Group A recorded 8 outcome events in 300 observations. Group B recorded 6 outcome events in 100 observations. Which group had the higher rate?

A. Group B, because 6/100 is a higher rate than 8/300; compare rates, not just counts. correct	**B.** Group A, because 300 observations is the larger group, so it must have the higher rate. choice
C. Group A, because 8 events is more than 6 events, so the raw count must decide the rate. choice	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, health, d2

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-052 | Level 2 | Answer D

Prompt. In a study about seat-zone airline boarding, the tracked outcome is average boarding time. Group A recorded 7 outcome events in 100 observations. Group B recorded 9 outcome events in 300 observations. Which group had the higher rate?

A. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**B.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice
C. Group B, because 9 events is more than 7 events, so the raw count must decide the rate. choice	**D.** Group A, because 7/100 is a higher rate than 9/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, travel, d2

A feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

B feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-053 | Level 2 | Answer B

Prompt. In a study about an online apartment repair portal, the tracked outcome is unresolved repair requests. Group A recorded 10 outcome events in 300 observations. Group B recorded 8 outcome events in 100 observations. Which group had the higher rate?

A. The raw counts are close enough that dividing by group size is unnecessary. choice	**B.** Group B, because 8/100 is a higher rate than 10/300; compare rates, not just counts. correct
C. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**D.** Group A, because 10 events is more than 8 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, housing, d2

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-054 | Level 2 | Answer C

Prompt. In a study about fruit placement in a school cafeteria line, the tracked outcome is fruit purchases. Group A recorded 9 outcome events in 100 observations. Group B recorded 11 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 11 events is more than 9 events, so the raw count must decide the rate. choice	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group A, because 9/100 is a higher rate than 11/300; compare rates, not just counts. correct	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, food, d2

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-055 | Level 2 | Answer D

Prompt. In a study about weekly call-center script coaching, the tracked outcome is customer complaint rates. Group A recorded 12 outcome events in 300 observations. Group B recorded 10 outcome events in 100 observations. Which group had the higher rate?

A. The raw counts are close enough that dividing by group size is unnecessary. choice	**B.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice
C. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** Group B, because 10/100 is a higher rate than 12/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, customer support, d2

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-056 | Level 2 | Answer A

Prompt. In a study about household water-leak alerts, the tracked outcome is average household water use. Group A recorded 11 outcome events in 100 observations. Group B recorded 13 outcome events in 300 observations. Which group had the higher rate?

A. Group A, because 11/100 is a higher rate than 13/300; compare rates, not just counts. correct	**B.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** Group B, because 13 events is more than 11 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, utilities, d2

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-057 | Level 2 | Answer B

Prompt. In a study about body-camera activation reminders, the tracked outcome is missing-camera-footage reports. Group A recorded 14 outcome events in 300 observations. Group B recorded 12 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 14 events is more than 12 events, so the raw count must decide the rate. choice	**B.** Group B, because 12/100 is a higher rate than 14/300; compare rates, not just counts. correct
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, governance, d2

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-058 | Level 2 | Answer C

Prompt. In a study about county mail-ballot tracking, the tracked outcome is phone calls asking ballot status. Group A recorded 4 outcome events in 100 observations. Group B recorded 6 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** Group B, because 6 events is more than 4 events, so the raw count must decide the rate. choice
C. Group A, because 4/100 is a higher rate than 6/300; compare rates, not just counts. correct	**D.** The raw counts are close enough that dividing by group size is unnecessary. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, election administration, d2

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-059 | Level 2 | Answer A

Prompt. In a study about language-learning spaced-review prompts, the tracked outcome is vocabulary retention quiz scores. Group A recorded 7 outcome events in 300 observations. Group B recorded 5 outcome events in 100 observations. Which group had the higher rate?

A. Group B, because 5/100 is a higher rate than 7/300; compare rates, not just counts. correct	**B.** Group A, because 7 events is more than 5 events, so the raw count must decide the rate. choice
C. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**D.** The raw counts are close enough that dividing by group size is unnecessary. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, learning, d2

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-060 | Level 2 | Answer C

Prompt. In a study about a beginner gym orientation, the tracked outcome is new-member cancellations. Group A recorded 6 outcome events in 100 observations. Group B recorded 8 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 8 events is more than 6 events, so the raw count must decide the rate. choice	**B.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice
C. Group A, because 6/100 is a higher rate than 8/300; compare rates, not just counts. correct	**D.** The raw counts are close enough that dividing by group size is unnecessary. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, fitness, d2

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-061 | Level 2 | Answer D

Prompt. In a study about safe-driving feedback for policyholders, the tracked outcome is reported minor crashes. Group A recorded 9 outcome events in 300 observations. Group B recorded 7 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 9 events is more than 7 events, so the raw count must decide the rate. choice	**B.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** Group B, because 7/100 is a higher rate than 9/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, insurance, d2

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-062 | Level 2 | Answer A

Prompt. In a study about top-of-article correction boxes, the tracked outcome is reader trust ratings. Group A recorded 8 outcome events in 100 observations. Group B recorded 10 outcome events in 300 observations. Which group had the higher rate?

A. Group A, because 8/100 is a higher rate than 10/300; compare rates, not just counts. correct	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group B, because 10 events is more than 8 events, so the raw count must decide the rate. choice	**D.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, media, d2

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

statistical_sense-063 | Level 2 | Answer B

Prompt. In a study about clearer farmers market price signs, the tracked outcome is price disputes. Group A recorded 11 outcome events in 300 observations. Group B recorded 9 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 11 events is more than 9 events, so the raw count must decide the rate. choice	**B.** Group B, because 9/100 is a higher rate than 11/300; compare rates, not just counts. correct
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** Group A, because 300 observations is the larger group, so it must have the higher rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, local commerce, d2

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

statistical_sense-064 | Level 2 | Answer D

Prompt. In a study about hotel quiet-hours text reminders, the tracked outcome is noise complaints from guest rooms. Group A recorded 10 outcome events in 100 observations. Group B recorded 12 outcome events in 300 observations. Which group had the higher rate?

A. The raw counts are close enough that dividing by group size is unnecessary. choice	**B.** Group B, because 12 events is more than 10 events, so the raw count must decide the rate. choice
C. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** Group A, because 10/100 is a higher rate than 12/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, hospitality, d2

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Level 3

Level 3 asks for steadier discrimination: the right answer is still objective, but the wrong answers often sound plausible.

statistical_sense-065 | Level 3 | Answer B

Prompt. In a study about evening bus service for late-shift workers, the tracked outcome is late-shift commute complaints. Group A recorded 7 outcome events in 100 observations. Group B recorded 9 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** Group A, because 7/100 is a higher rate than 9/300; compare rates, not just counts. correct
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, transportation, d3

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-066 | Level 3 | Answer A

Prompt. In a study about classroom phone lockers, the tracked outcome is classroom disruption reports. Group A recorded 10 outcome events in 300 observations. Group B recorded 8 outcome events in 100 observations. Which group had the higher rate?

A. Group B, because 8/100 is a higher rate than 10/300; compare rates, not just counts. correct	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group A, because 10 events is more than 8 events, so the raw count must decide the rate. choice	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, education, d3

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-067 | Level 3 | Answer D

Prompt. In a study about medication reminder texts for older patients, the tracked outcome is missed-dose reports. Group A recorded 9 outcome events in 100 observations. Group B recorded 11 outcome events in 300 observations. Which group had the higher rate?

A. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** Group A, because 9/100 is a higher rate than 11/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, health, d3

A feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-068 | Level 3 | Answer C

Prompt. In a study about a four-day workweek test for software engineers, the tracked outcome is staff resignations. Group A recorded 12 outcome events in 300 observations. Group B recorded 10 outcome events in 100 observations. Which group had the higher rate?

A. The raw counts are close enough that dividing by group size is unnecessary. choice	**B.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice
C. Group B, because 10/100 is a higher rate than 12/300; compare rates, not just counts. correct	**D.** Group A, because 12 events is more than 10 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, workplace, d3

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-069 | Level 3 | Answer B

Prompt. In a study about larger grocery unit-price labels, the tracked outcome is shopper overpayment complaints. Group A recorded 11 outcome events in 100 observations. Group B recorded 13 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 13 events is more than 11 events, so the raw count must decide the rate. choice	**B.** Group A, because 11/100 is a higher rate than 13/300; compare rates, not just counts. correct
C. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**D.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, retail, d3

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

D feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

statistical_sense-070 | Level 3 | Answer A

Prompt. In a study about late fines on children's books, the tracked outcome is library card renewals. Group A recorded 14 outcome events in 300 observations. Group B recorded 12 outcome events in 100 observations. Which group had the higher rate?

A. Group B, because 12/100 is a higher rate than 14/300; compare rates, not just counts. correct	**B.** Group A, because 14 events is more than 12 events, so the raw count must decide the rate. choice
C. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, public services, d3

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-071 | Level 3 | Answer D

Prompt. In a study about two-factor login for site administrators, the tracked outcome is admin account takeovers. Group A recorded 4 outcome events in 100 observations. Group B recorded 6 outcome events in 300 observations. Which group had the higher rate?

A. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**B.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** Group A, because 4/100 is a higher rate than 6/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, technology, d3

A feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

B feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-072 | Level 3 | Answer C

Prompt. In a study about brighter streetlights on Maple Street, the tracked outcome is reported thefts. Group A recorded 7 outcome events in 300 observations. Group B recorded 5 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 7 events is more than 5 events, so the raw count must decide the rate. choice	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group B, because 5/100 is a higher rate than 7/300; compare rates, not just counts. correct	**D.** Group A, because 300 observations is the larger group, so it must have the higher rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, public safety, d3

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

statistical_sense-073 | Level 3 | Answer B

Prompt. In a study about peer tutoring for first-year statistics students, the tracked outcome is course pass rates. Group A recorded 6 outcome events in 100 observations. Group B recorded 8 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 8 events is more than 6 events, so the raw count must decide the rate. choice	**B.** Group A, because 6/100 is a higher rate than 8/300; compare rates, not just counts. correct
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, education, d3

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

statistical_sense-074 | Level 3 | Answer A

Prompt. In a study about calorie labels on restaurant menus, the tracked outcome is dessert orders. Group A recorded 9 outcome events in 300 observations. Group B recorded 7 outcome events in 100 observations. Which group had the higher rate?

A. Group B, because 7/100 is a higher rate than 9/300; compare rates, not just counts. correct	**B.** Group A, because 9 events is more than 7 events, so the raw count must decide the rate. choice
C. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**D.** The raw counts are close enough that dividing by group size is unnecessary. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, nutrition, d3

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-075 | Level 3 | Answer C

Prompt. In a study about bank overdraft warning texts, the tracked outcome is overdraft fees. Group A recorded 8 outcome events in 100 observations. Group B recorded 10 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 10 events is more than 8 events, so the raw count must decide the rate. choice	**B.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice
C. Group A, because 8/100 is a higher rate than 10/300; compare rates, not just counts. correct	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, finance, d3

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-076 | Level 3 | Answer D

Prompt. In a study about timed-entry passes for weekend park trails, the tracked outcome is trail crowding complaints. Group A recorded 11 outcome events in 300 observations. Group B recorded 9 outcome events in 100 observations. Which group had the higher rate?

A. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** Group B, because 9/100 is a higher rate than 11/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, environment, d3

A feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-077 | Level 3 | Answer A

Prompt. In a study about free Friday museum admission, the tracked outcome is family visits. Group A recorded 10 outcome events in 100 observations. Group B recorded 12 outcome events in 300 observations. Which group had the higher rate?

A. Group A, because 10/100 is a higher rate than 12/300; compare rates, not just counts. correct	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group B, because 12 events is more than 10 events, so the raw count must decide the rate. choice	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, culture, d3

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-078 | Level 3 | Answer B

Prompt. In a study about youth soccer concussion checks, the tracked outcome is repeat head injuries. Group A recorded 13 outcome events in 300 observations. Group B recorded 11 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** Group B, because 11/100 is a higher rate than 13/300; compare rates, not just counts. correct
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** Group A, because 13 events is more than 11 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, sports, d3

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-079 | Level 3 | Answer C

Prompt. In a study about same-day donation receipts, the tracked outcome is donor retention. Group A recorded 12 outcome events in 100 observations. Group B recorded 14 outcome events in 300 observations. Which group had the higher rate?

A. The raw counts are close enough that dividing by group size is unnecessary. choice	**B.** Group B, because 14 events is more than 12 events, so the raw count must decide the rate. choice
C. Group A, because 12/100 is a higher rate than 14/300; compare rates, not just counts. correct	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, nonprofit, d3

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-080 | Level 3 | Answer D

Prompt. In a study about warehouse ergonomic training, the tracked outcome is back-injury claims. Group A recorded 6 outcome events in 300 observations. Group B recorded 4 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 6 events is more than 4 events, so the raw count must decide the rate. choice	**B.** Group A, because 300 observations is the larger group, so it must have the higher rate. choice
C. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**D.** Group B, because 4/100 is a higher rate than 6/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, workplace safety, d3

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

C feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-081 | Level 3 | Answer A

Prompt. In a study about apartment-lobby compost bins, the tracked outcome is landfill waste from participating buildings. Group A recorded 5 outcome events in 100 observations. Group B recorded 7 outcome events in 300 observations. Which group had the higher rate?

A. Group A, because 5/100 is a higher rate than 7/300; compare rates, not just counts. correct	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, environment, d3

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-082 | Level 3 | Answer B

Prompt. In a study about private-by-default app profiles, the tracked outcome is publicly visible profiles. Group A recorded 8 outcome events in 300 observations. Group B recorded 6 outcome events in 100 observations. Which group had the higher rate?

A. The raw counts are close enough that dividing by group size is unnecessary. choice	**B.** Group B, because 6/100 is a higher rate than 8/300; compare rates, not just counts. correct
C. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** Group A, because 8 events is more than 6 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, technology, d3

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-083 | Level 3 | Answer C

Prompt. In a study about same-day clinic appointment slots, the tracked outcome is patient no-show rates. Group A recorded 7 outcome events in 100 observations. Group B recorded 9 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group A, because 7/100 is a higher rate than 9/300; compare rates, not just counts. correct	**D.** Group B, because 9 events is more than 7 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, health, d3

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-084 | Level 3 | Answer D

Prompt. In a study about seat-zone airline boarding, the tracked outcome is average boarding time. Group A recorded 10 outcome events in 300 observations. Group B recorded 8 outcome events in 100 observations. Which group had the higher rate?

A. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**B.** Group A, because 10 events is more than 8 events, so the raw count must decide the rate. choice
C. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** Group B, because 8/100 is a higher rate than 10/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, travel, d3

A feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-085 | Level 3 | Answer B

Prompt. In a study about an online apartment repair portal, the tracked outcome is unresolved repair requests. Group A recorded 9 outcome events in 100 observations. Group B recorded 11 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 11 events is more than 9 events, so the raw count must decide the rate. choice	**B.** Group A, because 9/100 is a higher rate than 11/300; compare rates, not just counts. correct
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** Group B, because 300 observations is the larger group, so it must have the higher rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, housing, d3

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

statistical_sense-086 | Level 3 | Answer A

Prompt. In a study about fruit placement in a school cafeteria line, the tracked outcome is fruit purchases. Group A recorded 12 outcome events in 300 observations. Group B recorded 10 outcome events in 100 observations. Which group had the higher rate?

A. Group B, because 10/100 is a higher rate than 12/300; compare rates, not just counts. correct	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group A, because 12 events is more than 10 events, so the raw count must decide the rate. choice	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, food, d3

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-087 | Level 3 | Answer D

Prompt. In a study about weekly call-center script coaching, the tracked outcome is customer complaint rates. Group A recorded 11 outcome events in 100 observations. Group B recorded 13 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** Group B, because 13 events is more than 11 events, so the raw count must decide the rate. choice
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** Group A, because 11/100 is a higher rate than 13/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, customer support, d3

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-088 | Level 3 | Answer C

Prompt. In a study about household water-leak alerts, the tracked outcome is average household water use. Group A recorded 14 outcome events in 300 observations. Group B recorded 12 outcome events in 100 observations. Which group had the higher rate?

A. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group B, because 12/100 is a higher rate than 14/300; compare rates, not just counts. correct	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, utilities, d3

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-089 | Level 3 | Answer B

Prompt. In a study about body-camera activation reminders, the tracked outcome is missing-camera-footage reports. Group A recorded 4 outcome events in 100 observations. Group B recorded 6 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** Group A, because 4/100 is a higher rate than 6/300; compare rates, not just counts. correct
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** Group B, because 6 events is more than 4 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, governance, d3

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-090 | Level 3 | Answer A

Prompt. In a study about county mail-ballot tracking, the tracked outcome is phone calls asking ballot status. Group A recorded 7 outcome events in 300 observations. Group B recorded 5 outcome events in 100 observations. Which group had the higher rate?

A. Group B, because 5/100 is a higher rate than 7/300; compare rates, not just counts. correct	**B.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** Group A, because 7 events is more than 5 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, election administration, d3

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-091 | Level 3 | Answer D

Prompt. In a study about language-learning spaced-review prompts, the tracked outcome is vocabulary retention quiz scores. Group A recorded 6 outcome events in 100 observations. Group B recorded 8 outcome events in 300 observations. Which group had the higher rate?

A. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** Group A, because 6/100 is a higher rate than 8/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, learning, d3

A feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-092 | Level 3 | Answer C

Prompt. In a study about a beginner gym orientation, the tracked outcome is new-member cancellations. Group A recorded 9 outcome events in 300 observations. Group B recorded 7 outcome events in 100 observations. Which group had the higher rate?

A. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**B.** Group A, because 300 observations is the larger group, so it must have the higher rate. choice
C. Group B, because 7/100 is a higher rate than 9/300; compare rates, not just counts. correct	**D.** Group A, because 9 events is more than 7 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, fitness, d3

A feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

B feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-093 | Level 3 | Answer B

Prompt. In a study about safe-driving feedback for policyholders, the tracked outcome is reported minor crashes. Group A recorded 8 outcome events in 100 observations. Group B recorded 10 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 300 observations is the larger group, so it must have the higher rate. choice	**B.** Group A, because 8/100 is a higher rate than 10/300; compare rates, not just counts. correct
C. The raw counts are close enough that dividing by group size is unnecessary. choice	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, insurance, d3

A feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-094 | Level 3 | Answer A

Prompt. In a study about top-of-article correction boxes, the tracked outcome is reader trust ratings. Group A recorded 11 outcome events in 300 observations. Group B recorded 9 outcome events in 100 observations. Which group had the higher rate?

A. Group B, because 9/100 is a higher rate than 11/300; compare rates, not just counts. correct	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. There is no fair way to compare rates when group sizes differ, even if both counts are given. choice	**D.** Group A, because 11 events is more than 9 events, so the raw count must decide the rate. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, media, d3

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

D feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

statistical_sense-095 | Level 3 | Answer C

Prompt. In a study about clearer farmers market price signs, the tracked outcome is price disputes. Group A recorded 10 outcome events in 100 observations. Group B recorded 12 outcome events in 300 observations. Which group had the higher rate?

A. Group B, because 12 events is more than 10 events, so the raw count must decide the rate. choice	**B.** The raw counts are close enough that dividing by group size is unnecessary. choice
C. Group A, because 10/100 is a higher rate than 12/300; compare rates, not just counts. correct	**D.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, local commerce, d3

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

statistical_sense-096 | Level 3 | Answer D

Prompt. In a study about hotel quiet-hours text reminders, the tracked outcome is noise complaints from guest rooms. Group A recorded 13 outcome events in 300 observations. Group B recorded 11 outcome events in 100 observations. Which group had the higher rate?

A. The raw counts are close enough that dividing by group size is unnecessary. choice	**B.** There is no fair way to compare rates when group sizes differ, even if both counts are given. choice
C. Group A, because 300 observations is the larger group, so it must have the higher rate. choice	**D.** Group B, because 11/100 is a higher rate than 13/300; compare rates, not just counts. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, hospitality, d3

A feedback (Distractor): Not quite. Raw counts can mislead when group sizes differ. The fair comparison is the rate.

B feedback (Distractor): Not quite. Different group sizes are exactly when rates are useful; the counts and totals are enough to compare.

C feedback (Distractor): Not quite. A bigger group does not automatically have the higher rate. You still have to divide events by observations.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Level 4

Level 4 asks learners to hold more context in mind while applying the same method.

statistical_sense-097 | Level 4 | Answer D

Prompt. A report about evening bus service for late-shift workers says late-shift commute complaints rose from 8% to 13% after a test program. What is the clearest way to describe the change?

A. It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**B.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice
C. It rose by 5%, because a 5-point change and a 5% change are the same thing. choice	**D.** It rose by 5 percentage points, which is about a 63% increase compared with the starting number. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, transportation, d4

A feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

B feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

C feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-098 | Level 4 | Answer A

Prompt. A report about classroom phone lockers says classroom disruption reports fell from 14% to 9% after a test program. What is the clearest way to describe the change?

A. It fell by 5 percentage points, which is about a 36% decrease compared with the starting number. correct	**B.** It fell by 5%, because a 5-point change and a 5% change are the same thing. choice
C. It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**D.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, education, d4

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

C feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

D feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

statistical_sense-099 | Level 4 | Answer B

Prompt. A report about medication reminder texts for older patients says missed-dose reports rose from 10% to 15% after a test program. What is the clearest way to describe the change?

A. It rose by 15 percentage points, because the ending rate should be copied as the point change. choice	**B.** It rose by 5 percentage points, which is about a 50% increase compared with the starting number. correct
C. It rose by 5%, because a 5-point change and a 5% change are the same thing. choice	**D.** It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, health, d4

A feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

D feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

statistical_sense-100 | Level 4 | Answer C

Prompt. A report about a four-day workweek test for software engineers says staff resignations fell from 16% to 11% after a test program. What is the clearest way to describe the change?

A. It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**B.** It fell by 11 percentage points, because the ending rate should be copied as the point change. choice
C. It fell by 5 percentage points, which is about a 31% decrease compared with the starting number. correct	**D.** It fell by 5%, because a 5-point change and a 5% change are the same thing. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, workplace, d4

A feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

B feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

statistical_sense-101 | Level 4 | Answer D

Prompt. A report about larger grocery unit-price labels says shopper overpayment complaints rose from 12% to 17% after a test program. What is the clearest way to describe the change?

A. It rose by 5%, because a 5-point change and a 5% change are the same thing. choice	**B.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice
C. It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**D.** It rose by 5 percentage points, which is about a 42% increase compared with the starting number. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, retail, d4

A feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

B feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

C feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-102 | Level 4 | Answer A

Prompt. A report about late fines on children's books says library card renewals fell from 9% to 4% after a test program. What is the clearest way to describe the change?

A. It fell by 5 percentage points, which is about a 56% decrease compared with the starting number. correct	**B.** It fell by 4 percentage points, because the ending rate should be copied as the point change. choice
C. It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**D.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, public services, d4

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

C feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

D feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

statistical_sense-103 | Level 4 | Answer B

Prompt. A report about two-factor login for site administrators says admin account takeovers rose from 5% to 10% after a test program. What is the clearest way to describe the change?

A. It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**B.** It rose by 5 percentage points, which is about a 100% increase compared with the starting number. correct
C. It rose by 5%, because a 5-point change and a 5% change are the same thing. choice	**D.** It rose by 10 percentage points, because the ending rate should be copied as the point change. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, technology, d4

A feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

D feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

statistical_sense-104 | Level 4 | Answer C

Prompt. A report about brighter streetlights on Maple Street says reported thefts fell from 11% to 6% after a test program. What is the clearest way to describe the change?

A. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**B.** It fell by 6 percentage points, because the ending rate should be copied as the point change. choice
C. It fell by 5 percentage points, which is about a 45% decrease compared with the starting number. correct	**D.** It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, public safety, d4

A feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

B feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

statistical_sense-105 | Level 4 | Answer B

Prompt. A report about peer tutoring for first-year statistics students says course pass rates rose from 7% to 12% after a test program. What is the clearest way to describe the change?

A. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**B.** It rose by 5 percentage points, which is about a 71% increase compared with the starting number. correct
C. It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**D.** It rose by 12 percentage points, because the ending rate should be copied as the point change. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, education, d4

A feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

D feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

statistical_sense-106 | Level 4 | Answer C

Prompt. A report about calorie labels on restaurant menus says dessert orders fell from 13% to 8% after a test program. What is the clearest way to describe the change?

A. It fell by 5%, because a 5-point change and a 5% change are the same thing. choice	**B.** It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice
C. It fell by 5 percentage points, which is about a 38% decrease compared with the starting number. correct	**D.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, nutrition, d4

A feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

B feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

statistical_sense-107 | Level 4 | Answer A

Prompt. A report about bank overdraft warning texts says overdraft fees rose from 9% to 14% after a test program. What is the clearest way to describe the change?

A. It rose by 5 percentage points, which is about a 56% increase compared with the starting number. correct	**B.** It rose by 14 percentage points, because the ending rate should be copied as the point change. choice
C. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**D.** It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, finance, d4

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

C feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

D feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

statistical_sense-108 | Level 4 | Answer C

Prompt. A report about timed-entry passes for weekend park trails says trail crowding complaints fell from 15% to 10% after a test program. What is the clearest way to describe the change?

A. It fell by 10 percentage points, because the ending rate should be copied as the point change. choice	**B.** It fell by 5%, because a 5-point change and a 5% change are the same thing. choice
C. It fell by 5 percentage points, which is about a 33% decrease compared with the starting number. correct	**D.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, environment, d4

A feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

B feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

statistical_sense-109 | Level 4 | Answer A

Prompt. A report about free Friday museum admission says family visits rose from 11% to 16% after a test program. What is the clearest way to describe the change?

A. It rose by 5 percentage points, which is about a 45% increase compared with the starting number. correct	**B.** It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice
C. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**D.** It rose by 16 percentage points, because the ending rate should be copied as the point change. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, culture, d4

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

C feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

D feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

statistical_sense-110 | Level 4 | Answer D

Prompt. A report about youth soccer concussion checks says repeat head injuries fell from 17% to 12% after a test program. What is the clearest way to describe the change?

A. It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**B.** It fell by 12 percentage points, because the ending rate should be copied as the point change. choice
C. It fell by 5%, because a 5-point change and a 5% change are the same thing. choice	**D.** It fell by 5 percentage points, which is about a 29% decrease compared with the starting number. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, sports, d4

A feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

B feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

C feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-111 | Level 4 | Answer A

Prompt. A report about same-day donation receipts says donor retention rose from 4% to 9% after a test program. What is the clearest way to describe the change?

A. It rose by 5 percentage points, which is about a 125% increase compared with the starting number. correct	**B.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice
C. It rose by 5%, because a 5-point change and a 5% change are the same thing. choice	**D.** It rose by 9 percentage points, because the ending rate should be copied as the point change. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, nonprofit, d4

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

C feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

D feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

statistical_sense-112 | Level 4 | Answer C

Prompt. A report about warehouse ergonomic training says back-injury claims fell from 10% to 5% after a test program. What is the clearest way to describe the change?

A. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**B.** It fell by 5%, because a 5-point change and a 5% change are the same thing. choice
C. It fell by 5 percentage points, which is about a 50% decrease compared with the starting number. correct	**D.** It fell by 5 percentage points, because the ending rate should be copied as the point change. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, workplace safety, d4

A feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

B feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

statistical_sense-113 | Level 4 | Answer A

Prompt. A report about apartment-lobby compost bins says landfill waste from participating buildings rose from 6% to 11% after a test program. What is the clearest way to describe the change?

A. It rose by 5 percentage points, which is about a 83% increase compared with the starting number. correct	**B.** It rose by 11 percentage points, because the ending rate should be copied as the point change. choice
C. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**D.** It rose by 5%, because a 5-point change and a 5% change are the same thing. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, environment, d4

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

C feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

D feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

statistical_sense-114 | Level 4 | Answer C

Prompt. A report about private-by-default app profiles says publicly visible profiles fell from 12% to 7% after a test program. What is the clearest way to describe the change?

A. It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**B.** It fell by 7 percentage points, because the ending rate should be copied as the point change. choice
C. It fell by 5 percentage points, which is about a 42% decrease compared with the starting number. correct	**D.** It fell by 5%, because a 5-point change and a 5% change are the same thing. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, technology, d4

A feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

B feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

statistical_sense-115 | Level 4 | Answer C

Prompt. A report about same-day clinic appointment slots says patient no-show rates rose from 8% to 13% after a test program. What is the clearest way to describe the change?

A. It rose by 5%, because a 5-point change and a 5% change are the same thing. choice	**B.** It rose by 13 percentage points, because the ending rate should be copied as the point change. choice
C. It rose by 5 percentage points, which is about a 63% increase compared with the starting number. correct	**D.** It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, health, d4

A feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

B feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

statistical_sense-116 | Level 4 | Answer B

Prompt. A report about seat-zone airline boarding says average boarding time fell from 14% to 9% after a test program. What is the clearest way to describe the change?

A. It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**B.** It fell by 5 percentage points, which is about a 36% decrease compared with the starting number. correct
C. It fell by 5%, because a 5-point change and a 5% change are the same thing. choice	**D.** It fell by 9 percentage points, because the ending rate should be copied as the point change. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, travel, d4

A feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

D feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

statistical_sense-117 | Level 4 | Answer D

Prompt. A report about an online apartment repair portal says unresolved repair requests rose from 10% to 15% after a test program. What is the clearest way to describe the change?

A. It rose by 5%, because a 5-point change and a 5% change are the same thing. choice	**B.** It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice
C. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**D.** It rose by 5 percentage points, which is about a 50% increase compared with the starting number. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, housing, d4

A feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

B feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

C feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-118 | Level 4 | Answer A

Prompt. A report about fruit placement in a school cafeteria line says fruit purchases fell from 16% to 11% after a test program. What is the clearest way to describe the change?

A. It fell by 5 percentage points, which is about a 31% decrease compared with the starting number. correct	**B.** It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice
C. It fell by 5%, because a 5-point change and a 5% change are the same thing. choice	**D.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, food, d4

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

C feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

D feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

statistical_sense-119 | Level 4 | Answer B

Prompt. A report about weekly call-center script coaching says customer complaint rates rose from 12% to 17% after a test program. What is the clearest way to describe the change?

A. It rose by 5%, because a 5-point change and a 5% change are the same thing. choice	**B.** It rose by 5 percentage points, which is about a 42% increase compared with the starting number. correct
C. It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**D.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, customer support, d4

A feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

D feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

statistical_sense-120 | Level 4 | Answer C

Prompt. A report about household water-leak alerts says average household water use fell from 9% to 4% after a test program. What is the clearest way to describe the change?

A. It fell by 5%, because a 5-point change and a 5% change are the same thing. choice	**B.** It fell by 4 percentage points, because the ending rate should be copied as the point change. choice
C. It fell by 5 percentage points, which is about a 56% decrease compared with the starting number. correct	**D.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, utilities, d4

A feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

B feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

statistical_sense-121 | Level 4 | Answer B

Prompt. A report about body-camera activation reminders says missing-camera-footage reports rose from 5% to 10% after a test program. What is the clearest way to describe the change?

A. It rose by 5%, because a 5-point change and a 5% change are the same thing. choice	**B.** It rose by 5 percentage points, which is about a 100% increase compared with the starting number. correct
C. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**D.** It rose by 10 percentage points, because the ending rate should be copied as the point change. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, governance, d4

A feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

D feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

statistical_sense-122 | Level 4 | Answer D

Prompt. A report about county mail-ballot tracking says phone calls asking ballot status fell from 11% to 6% after a test program. What is the clearest way to describe the change?

A. It fell by 6 percentage points, because the ending rate should be copied as the point change. choice	**B.** It fell by 5%, because a 5-point change and a 5% change are the same thing. choice
C. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**D.** It fell by 5 percentage points, which is about a 45% decrease compared with the starting number. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, election administration, d4

A feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

B feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

C feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-123 | Level 4 | Answer B

Prompt. A report about language-learning spaced-review prompts says vocabulary retention quiz scores rose from 7% to 12% after a test program. What is the clearest way to describe the change?

A. It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**B.** It rose by 5 percentage points, which is about a 71% increase compared with the starting number. correct
C. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**D.** It rose by 12 percentage points, because the ending rate should be copied as the point change. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, learning, d4

A feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

D feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

statistical_sense-124 | Level 4 | Answer D

Prompt. A report about a beginner gym orientation says new-member cancellations fell from 13% to 8% after a test program. What is the clearest way to describe the change?

A. It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**B.** It fell by 8 percentage points, because the ending rate should be copied as the point change. choice
C. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**D.** It fell by 5 percentage points, which is about a 38% decrease compared with the starting number. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, fitness, d4

A feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

B feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

C feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-125 | Level 4 | Answer B

Prompt. A report about safe-driving feedback for policyholders says reported minor crashes rose from 9% to 14% after a test program. What is the clearest way to describe the change?

A. It rose by 14 percentage points, because the ending rate should be copied as the point change. choice	**B.** It rose by 5 percentage points, which is about a 56% increase compared with the starting number. correct
C. It rose by 5%, because a 5-point change and a 5% change are the same thing. choice	**D.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, insurance, d4

A feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

D feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

statistical_sense-126 | Level 4 | Answer D

Prompt. A report about top-of-article correction boxes says reader trust ratings fell from 15% to 10% after a test program. What is the clearest way to describe the change?

A. It fell by 10 percentage points, because the ending rate should be copied as the point change. choice	**B.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice
C. It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**D.** It fell by 5 percentage points, which is about a 33% decrease compared with the starting number. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, media, d4

A feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

B feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

C feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-127 | Level 4 | Answer A

Prompt. A report about clearer farmers market price signs says price disputes rose from 11% to 16% after a test program. What is the clearest way to describe the change?

A. It rose by 5 percentage points, which is about a 45% increase compared with the starting number. correct	**B.** It rose by 5%, because a 5-point change and a 5% change are the same thing. choice
C. It rose by 16 percentage points, because the ending rate should be copied as the point change. choice	**D.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, local commerce, d4

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

C feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

D feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

statistical_sense-128 | Level 4 | Answer D

Prompt. A report about hotel quiet-hours text reminders says noise complaints from guest rooms fell from 17% to 12% after a test program. What is the clearest way to describe the change?

A. It fell by 12 percentage points, because the ending rate should be copied as the point change. choice	**B.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice
C. It fell by 5%, because a 5-point change and a 5% change are the same thing. choice	**D.** It fell by 5 percentage points, which is about a 29% decrease compared with the starting number. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, hospitality, d4

A feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

B feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

C feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Level 5

Level 5 is for fluency: the learner should now name the thinking move and explain the elimination path.

statistical_sense-129 | Level 5 | Answer B

Prompt. A report about evening bus service for late-shift workers says late-shift commute complaints fell from 14% to 9% after a test program. What is the clearest way to describe the change?

A. It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**B.** It fell by 5 percentage points, which is about a 36% decrease compared with the starting number. correct
C. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**D.** It fell by 9 percentage points, because the ending rate should be copied as the point change. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, transportation, d5

A feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

D feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

statistical_sense-130 | Level 5 | Answer A

Prompt. A report about classroom phone lockers says classroom disruption reports rose from 10% to 15% after a test program. What is the clearest way to describe the change?

A. It rose by 5 percentage points, which is about a 50% increase compared with the starting number. correct	**B.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice
C. It rose by 15 percentage points, because the ending rate should be copied as the point change. choice	**D.** It rose by 5%, because a 5-point change and a 5% change are the same thing. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, education, d5

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

C feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

D feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

statistical_sense-131 | Level 5 | Answer D

Prompt. A report about medication reminder texts for older patients says missed-dose reports fell from 16% to 11% after a test program. What is the clearest way to describe the change?

A. It fell by 5%, because a 5-point change and a 5% change are the same thing. choice	**B.** It fell by 11 percentage points, because the ending rate should be copied as the point change. choice
C. It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**D.** It fell by 5 percentage points, which is about a 31% decrease compared with the starting number. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, health, d5

A feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

B feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

C feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-132 | Level 5 | Answer C

Prompt. A report about a four-day workweek test for software engineers says staff resignations rose from 12% to 17% after a test program. What is the clearest way to describe the change?

A. It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**B.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice
C. It rose by 5 percentage points, which is about a 42% increase compared with the starting number. correct	**D.** It rose by 5%, because a 5-point change and a 5% change are the same thing. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, workplace, d5

A feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

B feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

statistical_sense-133 | Level 5 | Answer B

Prompt. A report about larger grocery unit-price labels says shopper overpayment complaints fell from 9% to 4% after a test program. What is the clearest way to describe the change?

A. It fell by 5%, because a 5-point change and a 5% change are the same thing. choice	**B.** It fell by 5 percentage points, which is about a 56% decrease compared with the starting number. correct
C. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**D.** It fell by 4 percentage points, because the ending rate should be copied as the point change. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, retail, d5

A feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

D feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

statistical_sense-134 | Level 5 | Answer A

Prompt. A report about late fines on children's books says library card renewals rose from 5% to 10% after a test program. What is the clearest way to describe the change?

A. It rose by 5 percentage points, which is about a 100% increase compared with the starting number. correct	**B.** It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice
C. It rose by 5%, because a 5-point change and a 5% change are the same thing. choice	**D.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, public services, d5

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

C feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

D feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

statistical_sense-135 | Level 5 | Answer D

Prompt. A report about two-factor login for site administrators says admin account takeovers fell from 11% to 6% after a test program. What is the clearest way to describe the change?

A. It fell by 5%, because a 5-point change and a 5% change are the same thing. choice	**B.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice
C. It fell by 6 percentage points, because the ending rate should be copied as the point change. choice	**D.** It fell by 5 percentage points, which is about a 45% decrease compared with the starting number. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, technology, d5

A feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

B feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

C feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-136 | Level 5 | Answer C

Prompt. A report about brighter streetlights on Maple Street says reported thefts rose from 7% to 12% after a test program. What is the clearest way to describe the change?

A. It rose by 12 percentage points, because the ending rate should be copied as the point change. choice	**B.** It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice
C. It rose by 5 percentage points, which is about a 71% increase compared with the starting number. correct	**D.** It rose by 5%, because a 5-point change and a 5% change are the same thing. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, public safety, d5

A feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

B feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

statistical_sense-137 | Level 5 | Answer D

Prompt. A report about peer tutoring for first-year statistics students says course pass rates fell from 13% to 8% after a test program. What is the clearest way to describe the change?

A. It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**B.** It fell by 5%, because a 5-point change and a 5% change are the same thing. choice
C. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**D.** It fell by 5 percentage points, which is about a 38% decrease compared with the starting number. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, education, d5

A feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

B feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

C feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-138 | Level 5 | Answer C

Prompt. A report about calorie labels on restaurant menus says dessert orders rose from 9% to 14% after a test program. What is the clearest way to describe the change?

A. It rose by 14 percentage points, because the ending rate should be copied as the point change. choice	**B.** It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice
C. It rose by 5 percentage points, which is about a 56% increase compared with the starting number. correct	**D.** It rose by 5%, because a 5-point change and a 5% change are the same thing. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, nutrition, d5

A feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

B feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

statistical_sense-139 | Level 5 | Answer C

Prompt. A report about bank overdraft warning texts says overdraft fees fell from 15% to 10% after a test program. What is the clearest way to describe the change?

A. It fell by 10 percentage points, because the ending rate should be copied as the point change. choice	**B.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice
C. It fell by 5 percentage points, which is about a 33% decrease compared with the starting number. correct	**D.** It fell by 5%, because a 5-point change and a 5% change are the same thing. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, finance, d5

A feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

B feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

statistical_sense-140 | Level 5 | Answer D

Prompt. A report about timed-entry passes for weekend park trails says trail crowding complaints rose from 11% to 16% after a test program. What is the clearest way to describe the change?

A. It rose by 5%, because a 5-point change and a 5% change are the same thing. choice	**B.** It rose by 16 percentage points, because the ending rate should be copied as the point change. choice
C. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**D.** It rose by 5 percentage points, which is about a 45% increase compared with the starting number. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, environment, d5

A feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

B feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

C feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-141 | Level 5 | Answer A

Prompt. A report about free Friday museum admission says family visits fell from 17% to 12% after a test program. What is the clearest way to describe the change?

A. It fell by 5 percentage points, which is about a 29% decrease compared with the starting number. correct	**B.** It fell by 12 percentage points, because the ending rate should be copied as the point change. choice
C. It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**D.** It fell by 5%, because a 5-point change and a 5% change are the same thing. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, culture, d5

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

C feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

D feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

statistical_sense-142 | Level 5 | Answer B

Prompt. A report about youth soccer concussion checks says repeat head injuries rose from 4% to 9% after a test program. What is the clearest way to describe the change?

A. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**B.** It rose by 5 percentage points, which is about a 125% increase compared with the starting number. correct
C. It rose by 9 percentage points, because the ending rate should be copied as the point change. choice	**D.** It rose by 5%, because a 5-point change and a 5% change are the same thing. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, sports, d5

A feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

D feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

statistical_sense-143 | Level 5 | Answer C

Prompt. A report about same-day donation receipts says donor retention fell from 10% to 5% after a test program. What is the clearest way to describe the change?

A. It fell by 5%, because a 5-point change and a 5% change are the same thing. choice	**B.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice
C. It fell by 5 percentage points, which is about a 50% decrease compared with the starting number. correct	**D.** It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, nonprofit, d5

A feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

B feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

statistical_sense-144 | Level 5 | Answer D

Prompt. A report about warehouse ergonomic training says back-injury claims rose from 6% to 11% after a test program. What is the clearest way to describe the change?

A. It rose by 5%, because a 5-point change and a 5% change are the same thing. choice	**B.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice
C. It rose by 11 percentage points, because the ending rate should be copied as the point change. choice	**D.** It rose by 5 percentage points, which is about a 83% increase compared with the starting number. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, workplace safety, d5

A feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

B feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

C feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-145 | Level 5 | Answer A

Prompt. A report about apartment-lobby compost bins says landfill waste from participating buildings fell from 12% to 7% after a test program. What is the clearest way to describe the change?

A. It fell by 5 percentage points, which is about a 42% decrease compared with the starting number. correct	**B.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice
C. It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**D.** It fell by 5%, because a 5-point change and a 5% change are the same thing. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, environment, d5

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

C feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

D feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

statistical_sense-146 | Level 5 | Answer B

Prompt. A report about private-by-default app profiles says publicly visible profiles rose from 8% to 13% after a test program. What is the clearest way to describe the change?

A. It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**B.** It rose by 5 percentage points, which is about a 63% increase compared with the starting number. correct
C. It rose by 5%, because a 5-point change and a 5% change are the same thing. choice	**D.** It rose by 13 percentage points, because the ending rate should be copied as the point change. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, technology, d5

A feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

D feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

statistical_sense-147 | Level 5 | Answer A

Prompt. A report about same-day clinic appointment slots says patient no-show rates fell from 14% to 9% after a test program. What is the clearest way to describe the change?

A. It fell by 5 percentage points, which is about a 36% decrease compared with the starting number. correct	**B.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice
C. It fell by 9 percentage points, because the ending rate should be copied as the point change. choice	**D.** It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, health, d5

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

C feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

D feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

statistical_sense-148 | Level 5 | Answer B

Prompt. A report about seat-zone airline boarding says average boarding time rose from 10% to 15% after a test program. What is the clearest way to describe the change?

A. It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**B.** It rose by 5 percentage points, which is about a 50% increase compared with the starting number. correct
C. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**D.** It rose by 5%, because a 5-point change and a 5% change are the same thing. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, travel, d5

A feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

D feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

statistical_sense-149 | Level 5 | Answer D

Prompt. A report about an online apartment repair portal says unresolved repair requests fell from 16% to 11% after a test program. What is the clearest way to describe the change?

A. It fell by 5%, because a 5-point change and a 5% change are the same thing. choice	**B.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice
C. It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**D.** It fell by 5 percentage points, which is about a 31% decrease compared with the starting number. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, housing, d5

A feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

B feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

C feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-150 | Level 5 | Answer C

Prompt. A report about fruit placement in a school cafeteria line says fruit purchases rose from 12% to 17% after a test program. What is the clearest way to describe the change?

A. It rose by 17 percentage points, because the ending rate should be copied as the point change. choice	**B.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice
C. It rose by 5 percentage points, which is about a 42% increase compared with the starting number. correct	**D.** It rose by 5%, because a 5-point change and a 5% change are the same thing. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, food, d5

A feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

B feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

statistical_sense-151 | Level 5 | Answer B

Prompt. A report about weekly call-center script coaching says customer complaint rates fell from 9% to 4% after a test program. What is the clearest way to describe the change?

A. It fell by 4 percentage points, because the ending rate should be copied as the point change. choice	**B.** It fell by 5 percentage points, which is about a 56% decrease compared with the starting number. correct
C. It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**D.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, customer support, d5

A feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

D feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

statistical_sense-152 | Level 5 | Answer A

Prompt. A report about household water-leak alerts says average household water use rose from 5% to 10% after a test program. What is the clearest way to describe the change?

A. It rose by 5 percentage points, which is about a 100% increase compared with the starting number. correct	**B.** It rose by 10 percentage points, because the ending rate should be copied as the point change. choice
C. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**D.** It rose by 5%, because a 5-point change and a 5% change are the same thing. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, utilities, d5

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

C feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

D feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

statistical_sense-153 | Level 5 | Answer D

Prompt. A report about body-camera activation reminders says missing-camera-footage reports fell from 11% to 6% after a test program. What is the clearest way to describe the change?

A. It fell by 5%, because a 5-point change and a 5% change are the same thing. choice	**B.** It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice
C. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**D.** It fell by 5 percentage points, which is about a 45% decrease compared with the starting number. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, governance, d5

A feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

B feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

C feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-154 | Level 5 | Answer C

Prompt. A report about county mail-ballot tracking says phone calls asking ballot status rose from 7% to 12% after a test program. What is the clearest way to describe the change?

A. It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**B.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice
C. It rose by 5 percentage points, which is about a 71% increase compared with the starting number. correct	**D.** It rose by 5%, because a 5-point change and a 5% change are the same thing. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, election administration, d5

A feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

B feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

statistical_sense-155 | Level 5 | Answer A

Prompt. A report about language-learning spaced-review prompts says vocabulary retention quiz scores fell from 13% to 8% after a test program. What is the clearest way to describe the change?

A. It fell by 5 percentage points, which is about a 38% decrease compared with the starting number. correct	**B.** It fell by 5%, because a 5-point change and a 5% change are the same thing. choice
C. It fell by 8 percentage points, because the ending rate should be copied as the point change. choice	**D.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, learning, d5

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

C feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

D feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

statistical_sense-156 | Level 5 | Answer C

Prompt. A report about a beginner gym orientation says new-member cancellations rose from 9% to 14% after a test program. What is the clearest way to describe the change?

A. It rose by 14 percentage points, because the ending rate should be copied as the point change. choice	**B.** It cannot be described without knowing the total population, even though the starting and ending rates are given. choice
C. It rose by 5 percentage points, which is about a 56% increase compared with the starting number. correct	**D.** It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is C because it reads the quantity in the right context.

Tags: statistical_sense, statistics, fitness, d5

A feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

B feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

C feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

D feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

statistical_sense-157 | Level 5 | Answer B

Prompt. A report about safe-driving feedback for policyholders says reported minor crashes fell from 15% to 10% after a test program. What is the clearest way to describe the change?

A. It fell by 5%, because a 5-point change and a 5% change are the same thing. choice	**B.** It fell by 5 percentage points, which is about a 33% decrease compared with the starting number. correct
C. It fell by 10 percentage points, because the ending rate should be copied as the point change. choice	**D.** It fell by about 5% compared with the ending number, because the final rate is the only number that matters. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, insurance, d5

A feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

D feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

statistical_sense-158 | Level 5 | Answer A

Prompt. A report about top-of-article correction boxes says reader trust ratings rose from 11% to 16% after a test program. What is the clearest way to describe the change?

A. It rose by 5 percentage points, which is about a 45% increase compared with the starting number. correct	**B.** It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice
C. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**D.** It rose by 16 percentage points, because the ending rate should be copied as the point change. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is A because it reads the quantity in the right context.

Tags: statistical_sense, statistics, media, d5

A feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

B feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

C feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

D feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

statistical_sense-159 | Level 5 | Answer D

Prompt. A report about clearer farmers market price signs says price disputes fell from 17% to 12% after a test program. What is the clearest way to describe the change?

A. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**B.** It fell by 12 percentage points, because the ending rate should be copied as the point change. choice
C. It fell by 5%, because a 5-point change and a 5% change are the same thing. choice	**D.** It fell by 5 percentage points, which is about a 29% decrease compared with the starting number. correct

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is D because it reads the quantity in the right context.

Tags: statistical_sense, statistics, local commerce, d5

A feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

B feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.

C feedback (Distractor): Not quite. Percentage points and percent change are different; the starting value matters for percent change.

D feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

statistical_sense-160 | Level 5 | Answer B

Prompt. A report about hotel quiet-hours text reminders says noise complaints from guest rooms rose from 4% to 9% after a test program. What is the clearest way to describe the change?

A. It cannot be described without knowing the total population, even though the starting and ending rates are given. choice	**B.** It rose by 5 percentage points, which is about a 125% increase compared with the starting number. correct
C. It rose by about 5% compared with the ending number, because the final rate is the only number that matters. choice	**D.** It rose by 9 percentage points, because the ending rate should be copied as the point change. choice

Core explanation. For numbers, pay attention to group size and to the difference between percentage points and percent change.

Process commentary. To uncover the answer, identify the denominator, baseline, rate, or comparison that makes the number meaningful. The correct choice is B because it reads the quantity in the right context.

Tags: statistical_sense, statistics, hospitality, d5

A feedback (Distractor): Not quite. The starting and ending rates are enough to describe this change.

B feedback (Correct): Correct. This answer reads the numbers with the right comparison. For numbers, pay attention to group size and to the difference between percentage points and percent change.

C feedback (Distractor): Not quite. Percent change compares the movement with the starting value, not only the final value.

D feedback (Distractor): Not quite. Percentage points are found by subtracting the starting rate from the ending rate, not by copying the ending value.